

Meissner Minimality in the Confocal Peabody Family

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Abstract

Arelio, Montejano, and Oliveros introduced a three-parameter family of constant-width bodies obtained by replacing the six edge neighborhoods of a regular Reuleaux tetrahedron with envelopes of spheres guided by confocal quadrics. The family contains both classical Meissner bodies as a circle-line degeneration and Robert’s tetrahedrally symmetric body as a parabolic limit. We prove, within this regular-tetrahedron confocal family, that the Meissner degenerations minimize volume. The geometric step is an exact normal-sphere decomposition: for width one,

$$\text{Vol } K(\mathbf{e}, \sigma) = V_M + \Phi(e_1) + \Phi(e_2) + \Phi(e_3),$$

independently of the eight orientation choices. A canonical confocal chart reduces the one-pair contribution to a fixed one-dimensional integral. Outward-rounded interval arithmetic then proves $\Phi(e) > 0$ for every $0 < e < 1$, using three complementary certificates near $e = 0$, on a compact bulk interval, and near the parabolic endpoint. Equality occurs exactly when all three parameters vanish. The computer-assisted part uses 2,590 terminal interval boxes. The result is restricted to regular-tetrahedron confocal Peabodies and does not resolve the three-dimensional Blaschke–Lebesgue problem.

Status of this draft

This manuscript is an expository working draft based on the frozen package **v1.0.0-certified** [14]. The scalar interval certificates have passed the archived internal validation. Before submission or external circulation as a final paper, the normal-sphere decomposition, the one-pair derivation, and the factor-of-eight normalization should complete the independent release audit. No claim in this draft extends beyond regular-tetrahedron confocal Peabodies.

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1 Introduction

A convex body in $\mathbb{R}^3$ has constant width d when the distance between the two supporting planes orthogonal to any direction is always d . Spheres are the most familiar examples, but constant-width bodies need not be rotationally symmetric and need not be smooth. The classical two-dimensional Blaschke–Lebesgue theorem identifies the Reuleaux triangle as the minimum-area body of prescribed width [4, 10, 7]. In dimension three, the corresponding minimum-volume problem remains open. The two noncongruent Meissner bodies are conjectured to be global minimizers [9, 11, 13].

For width one their common volume is

$$V_M = \pi \left(\frac{2}{3} - \frac{\sqrt{3}}{4} \arccos \frac{1}{3} \right) = 0.419860045965080 \dots \quad (1.1)$$

The best universal lower bounds are strictly smaller; recent certified work gives a coefficient above 0.411040473721188 [8], while the conjectural upper endpoint remains (1.1). The theorem proved here does not narrow that universal interval. Instead, it completely resolves one explicit and geometrically natural family.

Arelío, Montejano, and Oliveros introduced *Peabodies*, obtained from the Reuleaux tetrahedron by replacing neighborhoods of all six edges with sections of envelopes of spheres. Their construction is guided by confocal quadrics and produces genuine bodies of constant width. The family contains the two Meissner bodies and Robert’s tetrahedrally symmetric body, and the source exhibits a continuous deformation between them [2]. The construction is especially attractive for optimization because feasibility is built into the geometry: every admissible parameter choice is already a constant-width body.

We restrict throughout to the regular-tetrahedron confocal family. The regular tetrahedron has three pairs of opposite edges. Each pair receives an elliptic-hyperbolic confocal device with a parameter $e_i \in [0, 1)$, and a bit $\sigma_i \in \{0, 1\}$ records which of the two edges carries which device. We write

$$\mathbf{e} = (e_1, e_2, e_3), \quad \boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3).$$

The value $e_i = 0$ is the circle-line Meissner degeneration; the limit $e_i \rightarrow 1^-$ is parabolic.

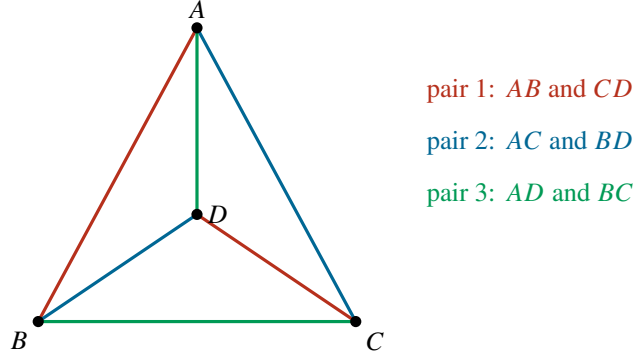


Figure 1: A projected regular tetrahedron with its three pairs of opposite edges. Each pair contributes one independent scalar parameter after the normal-sphere decomposition.

Theorem 1.1 (Main theorem). *Let $K(\mathbf{e}, \sigma)$ be any width-one regular-tetrahedron confocal Peabody with*

$$\mathbf{e} \in [0, 1]^3, \quad \sigma \in \{0, 1\}^3.$$

Then

$$\text{Vol } K(\mathbf{e}, \sigma) \geq V_M. \tag{1.2}$$

Equality holds if and only if

$$e_1 = e_2 = e_3 = 0.$$

Within this family, the equality cases are the two classical Meissner degenerations.

The proof has two conceptually separate parts.

1. **Exact geometry.** A normal-sphere partition eliminates the four spherical caps and yields an exact additive decomposition

$$\text{Vol } K(\mathbf{e}, \sigma) = V_M + \Phi(e_1) + \Phi(e_2) + \Phi(e_3), \tag{1.3}$$

independent of σ . A canonical confocal chart gives a fixed one-dimensional integral for the scalar function Φ .

2. **Certified scalar positivity.** A computer-assisted proof establishes

$$\Phi(e) > 0 \quad (0 < e < 1), \tag{1.4}$$

while $\Phi(0) = 0$. The interval is covered by a local endpoint certificate, a compact bulk certificate, and a stable tail certificate.

The exact decomposition (1.3) is the structural reason the problem is finite. A three-parameter, eight-orientation volume comparison reduces to one scalar inequality. The interval proof does not require global monotonicity of Φ , and in fact no global sign claim is made for Φ' .

The organization is as follows. [Section 2](#) recalls the source construction. [Section 3](#) proves the exact opposite-pair decomposition and orientation independence. [Section 4](#) derives the one-pair integral and the stable q -chart. [Section 5](#) describes the local, bulk, and tail certificates. [Section 6](#) assembles the theorem and equality cases. [Section 7](#) records the proof authority and trust boundary. [Section 8](#) states the precise scope and explains why no semi-regular-tetrahedron extension is implicit in the argument. The appendices expand the calculations at a level intended to be readable, where possible, by an undergraduate who has completed multivariable calculus.

2 Peabody geometry and the confocal construction

We work first at the native scale of the Peabody construction: a regular tetrahedron $ABCD$ with side length two. The resulting body has constant width two. The width-one normalization is applied only after the width-two surface and volume formulas have been formed.

2.1 Confocal pea-pod devices

A pea-pod device is a one-parameter family of balls. The centers move along a conic—an ellipse, a hyperbola, or a parabola—and the radius changes so that each moving ball remains tangent to two fixed circles in the device plane. The center curve is called the *pea string*. Two devices are confocal when they share an axis, lie in orthogonal planes, and their pea strings are confocal quadrics in the sense of Arelio, Montejano, and Oliveros [2].

The elementary geometric input is the following distance identity.

Theorem 2.1 (Confocal distance-plus-radii identity [2]). *Let x and y be centers on a pair of confocal pea strings, and let $R(x)$ and $R(y)$ be the corresponding ball radii. Then*

$$|x - y| + R(x) + R(y) = L, \quad (2.1)$$

where L is independent of x and y . When the longitudinal beams are opposite edges of a regular tetrahedron of side two, $L = 2$.

The source derives (2.1) from a classical alternating-distance identity for confocal quadrics. In the canonical coordinates used later, the identity becomes an immediate algebraic equality; see Section B.

2.2 Wedge-pod surfaces

For a fixed confocal pair, choose one center x on the first pea string and one center y on the second. On the line through x and y , mark the two boundary points

$$U_1 = x + R(x)n, \quad U_2 = y - R(y)n, \quad n = \frac{x - y}{|x - y|}.$$

Equation (2.1) gives

$$|U_1 - U_2| = L.$$

As x and y vary, the two maps sweep out a pair of smooth surfaces with boundary, called the *wedge-pod surfaces*. The joining segment is orthogonal to the tangent plane at both endpoints. Thus the two patches are paired by length- L binormals.

For the regular tetrahedron there are three pairs of opposite edges. Applying one confocal pair to each opposite-edge pair produces six wedge-pod surfaces. Four radius-two spherical caps close the remaining vertex regions. The source proves that these ten patches assemble to the boundary of a convex body of constant width two.

Theorem 2.2 (Peabody assembly theorem [2]). *The six wedge-pod surfaces and four spherical caps associated with three admissible confocal pairs on the opposite edges of a regular tetrahedron of side two form the boundary of a convex body of constant width two. The boundary is smooth except at the four tetrahedral vertices.*

This theorem is the feasibility input for the present paper. We do not need to prove constant width numerically. Every member of the family is legal before any volume calculation is performed.

2.3 Parameters and limiting cases

For one elliptic-hyperbolic pair, let

$$e = \frac{c}{a} \in [0, 1)$$

be the eccentricity-type parameter of the canonical chart. The endpoints have the following interpretation.

- $e = 0$: the ellipse becomes a circle and the hyperbola becomes an orthogonal line. The wedge-pod surfaces reduce to the classical Meissner edge surgery.
- $0 < e < 1$: the construction is genuinely elliptic-hyperbolic.
- $e \rightarrow 1^-$: the conics approach the parabolic regime associated with Robert's body.

The source explicitly identifies the circle-line degeneration with the Meissner construction and gives a continuous deformation through Peabodies from Robert's body to the Meissner bodies [2]. The regular-tetrahedron family therefore supplies a natural test of whether confocal edge surgery can reduce the Meissner volume.

For each of the three opposite-edge pairs, an orientation bit records which physical edge receives the elliptic device. We denote the resulting width-two body by

$$\tilde{K}(\mathbf{e}, \sigma),$$

and the width-one rescaling by

$$K(\mathbf{e}, \sigma) = \frac{1}{2} \tilde{K}(\mathbf{e}, \sigma).$$

The next section shows that the orientation bits disappear from the exact area and volume formulas.

3 Exact opposite-pair volume decomposition

The four spherical caps depend on all three opposite-edge pairs through their boundary curves. At first sight this seems to couple the three parameters. The coupling disappears on the unit normal sphere.

3.1 Reverse Gauss maps and antipodal support points

A constant-width body is strictly convex, so for each $n \in \mathbb{S}^2$ there is a unique support point $R_K(n)$ whose outward supporting normal is n . The map R_K is the reverse Gauss map.

Lemma 3.1 (Opposite support point). *If K has constant width two, then*

$$R_K(-n) = R_K(n) - 2n \quad (n \in \mathbb{S}^2). \quad (3.1)$$

Proof. Put $x = R_K(n)$ and $y = R_K(-n)$. The two supporting planes have separation two, hence

$$(x - y) \cdot n = 2.$$

The diameter of a width-two body is two, so $|x - y| \leq 2$. Cauchy–Schwarz gives

$$2 = (x - y) \cdot n \leq |x - y| \leq 2.$$

Equality throughout forces $x - y = 2n$, which is (3.1). □

Let C_A, C_B, C_C, C_D be the four radius-two spherical caps, and let

$$\Omega_A, \Omega_B, \Omega_C, \Omega_D \subset \mathbb{S}^2$$

be their Gauss images. For a point $x \in C_A$, the outward normal is

$$n = \frac{x - A}{2}.$$

By [Theorem 3.1](#), the opposite support point is A . Conversely, every support normal at A pairs with a point of the cap. Thus the full normal cone at A is exactly

$$N_K(A) = -\Omega_A. \quad (3.2)$$

The same statement holds at all four vertices.

3.2 Gauss images of opposite wedge-pod surfaces

Write the two wedge-pod surfaces associated with opposite-edge pair i as

$$W_i^+, \quad W_i^-.$$

The source binormal pairing joins every smooth point of W_i^+ to a point of W_i^- by a length-two segment orthogonal to both tangent planes. The outward normals are opposite. If

$$\Gamma_i = \text{Gauss}(W_i^+),$$

then

$$\text{Gauss}(W_i^-) = -\Gamma_i. \quad (3.3)$$

The Peabody boundary is smooth away from the four tetrahedral vertices. The patch seams therefore contribute only spherical measure zero. The smooth-patch Gauss images and the four vertex normal cones partition $\mathbb{S}^2$. Using (3.2) and (3.3),

$$4\pi = 2 \sum_{v \in \{A, B, C, D\}} |\Omega_v| + 2 \sum_{i=1}^3 |\Gamma_i|. \quad (3.4)$$

Hence

$$\sum_v |\Omega_v| = 2\pi - \sum_{i=1}^3 |\Gamma_i|. \quad (3.5)$$

A spherical patch on a sphere of radius two has physical area four times the area of its Gauss image. Therefore the total area of the four caps is

$$\sum_v |C_v| = 4 \sum_v |\Omega_v| = 8\pi - 4 \sum_{i=1}^3 |\Gamma_i|. \quad (3.6)$$

Adding the six wedge-pod areas gives the exact width-two surface formula

$$S_2 = 8\pi + \sum_{i=1}^3 (|W_i^+| + |W_i^-| - 4|\Gamma_i|). \quad (3.7)$$

3.3 One scalar per opposite-edge pair

Define

$$\Psi_i := |W_i^+| + |W_i^-| - 4|\Gamma_i|. \quad (3.8)$$

All three opposite-edge configurations of the regular tetrahedron are congruent. Moreover, within a fixed pair, exchanging the elliptic and hyperbolic devices merely exchanges the two physical wedge surfaces. Their area sum is unchanged, and their Gauss images are antipodal with equal area. Thus there is a single scalar function $\Psi(e)$, independent of the orientation bit, such that

$$S_2 = 8\pi + \Psi(e_1) + \Psi(e_2) + \Psi(e_3). \quad (3.9)$$

The argument is initially written for $0 < e_i < 1$, where all patch parametrizations are nondegenerate. The explicit one-pair formula in [Section 4](#) extends continuously to $e_i = 0$, so (3.9) also holds at the Meissner degenerations.

Remark 3.2 (Relation to normal-sphere formulas for Meissner polyhedra). The bookkeeping in (3.4) is analogous to the linked-rectangle normal-sphere decomposition used by Bogosel for Meissner polyhedra [5]. Here the local edge pieces are not circular wedge/spindle pairs, but the antipodal Gauss-image mechanism is the same.

3.4 Width normalization and the additive volume formula

For a three-dimensional body of constant width d , Blaschke's identity is

$$V = \frac{d}{2}S - \frac{\pi}{3}d^3. \quad (3.10)$$

See, for example, [9, 1, 5]. At width two, (3.10) and (3.9) give

$$V_2 = S_2 - \frac{8\pi}{3} = \frac{16\pi}{3} + \sum_{i=1}^3 \Psi(e_i). \quad (3.11)$$

Scaling lengths by one half changes volume by 1/8, so the width-one volume is

$$V_1 = \frac{2\pi}{3} + \frac{1}{8} \sum_{i=1}^3 \Psi(e_i). \quad (3.12)$$

At the all-zero degeneration,

$$V_M = \frac{2\pi}{3} + \frac{3\Psi(0)}{8}. \quad (3.13)$$

Define the one-pair width-one increment

$$\Phi(e) := \frac{\Psi(e) - \Psi(0)}{8}. \quad (3.14)$$

Subtracting (3.13) from (3.12) yields

Proposition 3.3 (Exact additive reduction). *For every admissible regular-tetrahedron confocal Peabody,*

$$\boxed{\text{Vol } K(\mathbf{e}, \sigma) = V_M + \Phi(e_1) + \Phi(e_2) + \Phi(e_3)}. \quad (3.15)$$

The right-hand side is independent of σ .

The factor 1/8 in (3.14) is a volume scaling factor. There is no additional surface-area scaling after Blaschke's identity has already converted the width-two surface formula into the width-two volume formula. This normalization is reviewed line by line in [Section A](#) and in the independent release-audit materials.

4 The one-pair formula and the stable scalar chart

We now derive the scalar $\Psi(e)$ for one opposite-edge pair. The calculation is exact. Numerical quadrature appears only later as audit evidence; it is not used to define the formula.

4.1 Canonical elliptic-hyperbolic coordinates

Fix an orthonormal frame $(\mathbf{k}, \mathbf{p}, \mathbf{q})$. Let

$$e = \frac{c}{a} \in [0, 1), \quad b = a\sqrt{1 - e^2}. \quad (4.1)$$

The regular-tetrahedron beam constraints give

$$b^2 = \frac{3 + 3e^2 + 4\sqrt{2}e}{1 - e^2}, \quad a = \frac{b}{\sqrt{1 - e^2}}. \quad (4.2)$$

A derivation from the beam endpoint equation is given in [Section B](#).

Define

$$\theta = \arccos \frac{1}{b}, \quad \eta = \operatorname{arsinh} \frac{1}{b}, \quad u_0 = \sin \theta, \quad v_0 = \cosh \eta. \quad (4.3)$$

The center curves are

$$X(t) = a \sin t \mathbf{k} + b \cos t \mathbf{p}, \quad \theta \leq t \leq \pi - \theta, \quad (4.4)$$

$$Y(\xi) = ae \cosh \xi \mathbf{k} + b \sinh \xi \mathbf{q}, \quad -\eta \leq \xi \leq \eta. \quad (4.5)$$

Set

$$D = \cosh \xi - e \sin t. \quad (4.6)$$

A direct expansion yields

$$d := |X - Y| = aD. \quad (4.7)$$

The two pea radii are

$$R_E = ae(\sin t - u_0), \quad R_H = a(v_0 - \cosh \xi). \quad (4.8)$$

The regular beam relation also implies

$$a(v_0 - eu_0) = 2, \quad (4.9)$$

so

$$d + R_E + R_H = 2. \quad (4.10)$$

This is [Theorem 2.1](#) in the canonical chart.

4.2 Conformal normal coordinates

The unit vector from the hyperbolic center to the elliptic center is

$$n(t, \xi) = \frac{(\sin t - e \cosh \xi) \mathbf{k} + \sqrt{1 - e^2} \cos t \mathbf{p} - \sqrt{1 - e^2} \sinh \xi \mathbf{q}}{D}. \quad (4.11)$$

Differentiation gives the conformal identities

$$n_t \cdot n_\xi = 0, \quad |n_t| = |n_\xi| = \frac{b}{d}. \quad (4.12)$$

Thus the spherical area element of the Gauss image is

$$d\omega = \frac{b^2}{d^2} dt d\xi. \quad (4.13)$$

The coordinate differentiation is expanded in [Section C](#).

The two paired boundary points are

$$U_E = X + R_E n, \quad U_H = Y - R_H n. \quad (4.14)$$

Using $X - Y = dn$ and differentiating (4.10), the normal components cancel and one obtains

$$(U_E)_t = (2 - R_H)n_t, \quad (U_E)_\xi = R_E n_\xi, \quad (4.15)$$

$$(U_H)_t = -R_H n_t, \quad (U_H)_\xi = -(2 - R_E)n_\xi. \quad (4.16)$$

Therefore

$$|W_E| = \iint R_E (2 - R_H) d\omega, \quad (4.17)$$

$$|W_H| = \iint R_H (2 - R_E) d\omega, \quad (4.18)$$

$$|\Gamma| = \iint d\omega. \quad (4.19)$$

Substituting into (3.8) and using $R_E + R_H = 2 - d$ gives the collapsed two-dimensional formula

$$\boxed{\Psi(e) = -2b^2 \int_\theta^{\pi-\theta} \int_{-\eta}^\eta \frac{d + R_E R_H}{d^2} d\xi dt.} \quad (4.20)$$

4.3 Reduction to one dimension

For fixed t , set

$$A = \sin t, \quad C = eA, \quad y = \tanh \frac{\eta}{2}.$$

The elementary kernel

$$I(C) := \int_{-\eta}^\eta \frac{d\xi}{\cosh \xi - C} = \frac{4}{\sqrt{1 - C^2}} \arctan \left(y \sqrt{\frac{1 + C}{1 - C}} \right) \quad (4.21)$$

results from the substitution $z = \tanh(\xi/2)$. A derivative identity for I , together with $\sinh \eta = 1/b$, reduces (4.20) to

$$\boxed{\Psi(e) = -4b^2 \int_\theta^{\pi/2} \left[\frac{I(e \sin t)}{a} + \frac{e(\sin t - u_0)}{1 - e^2 \sin^2 t} \left((v_0 e \sin t - 1) I(e \sin t) + \frac{2}{b} \right) \right] dt.} \quad (4.22)$$

The derivation is supplied in [Section D](#).

At $e = 0$,

$$a = b = \sqrt{3}, \quad I(0) = \frac{\pi}{3}, \quad \pi - 2\theta = \arccos \frac{1}{3},$$

and hence

$$\boxed{\Psi(0) = -\frac{2\pi}{\sqrt{3}} \arccos \frac{1}{3}.} \quad (4.23)$$

Combining (4.23) with (3.13) recovers the exact volume (1.1).

4.4 Fixed interval and stable q -chart

The substitution

$$x = b \cos t$$

removes the moving lower endpoint and produces

$$\Psi(e) = \int_0^1 H(e, x) \, dx. \quad (4.24)$$

For certification, the most stable parameter is

$$q = \frac{1-e}{1+e}, \quad e = \frac{1-q}{1+q}. \quad (4.25)$$

Thus $e = 0$ corresponds to $q = 1$, and $e \rightarrow 1^-$ corresponds to $q \rightarrow 0^+$. Put

$$\kappa = 1 + \sqrt{2}, \quad K = \kappa^2 = 3 + 2\sqrt{2}. \quad (4.26)$$

For $0 \leq q \leq 1$ and $0 \leq x \leq 1$, define

$$D = \sqrt{K^2 + q^2}, \quad (4.27)$$

$$R = \sqrt{K^2 + q^2 - 2Kqx^2}, \quad (4.28)$$

$$T = \sqrt{2K(K^2 + q^2) + K^2(1-q)^2x^2}, \quad (4.29)$$

$$\delta = \frac{2K(1-x^2)}{R + K - q}, \quad (4.30)$$

$$M = \frac{K(q - 2Kx^2)}{R + K} + (1-K)R - K^2 - qR - q - q^2. \quad (4.31)$$

Set

$$P_1 = -\frac{16\sqrt{2}\kappa(K^2 + q^2)}{RT}, \quad (4.32)$$

$$P_2 = -\frac{4K(1-q^2)(K^2 + q^2)\delta M}{RT^3}, \quad (4.33)$$

$$Q_0 = -\frac{4K(1-q^2)(K^2 + q^2)\delta}{RT^2}, \quad (4.34)$$

$$Z = \frac{K((1+q)D + (1-q)R)}{T(K + q + D)}. \quad (4.35)$$

Then the fixed-interval integrand is

$$H(q, x) = (P_1 + P_2) \arctan Z + Q_0, \quad (4.36)$$

and

$$\Psi(e(q)) = \int_0^1 H(q, x) \, dx. \quad (4.37)$$

The formula extends continuously to $q = 0$. The algebraic conversion from (4.22) to (4.36) is recorded in [Section E](#); independent Python and Mathematica implementations agree to high precision on the archived audit grid.



Figure 2: Exact decomposition of the scalar parameter interval. The drawing is not to scale near the two short endpoint ranges.

5 Certified positivity of the scalar increment

The exact family reduction in [Theorem 3.3](#) leaves one task:

$$\Phi(e) > 0 \quad (0 < e < 1).$$

The proof does not establish global monotonicity of Φ , and it does not require the integrand $H(q, x)$ to be pointwise positive. The target is the integrated quantity.

5.1 Certificate architecture

The interval $[0, 1)$ is divided into three exact pieces:

$$\left[0, \frac{1}{100}\right], \quad \left[\frac{1}{100}, \frac{199}{200}\right], \quad \left[\frac{199}{200}, 1\right). \quad (5.1)$$

The first range is handled through a positive derivative bound, the middle range by direct positivity of Φ , and the tail by a stable direct enclosure in the q -chart.

All interval endpoints are rational. The central and tail charts meet exactly because

$$e = \frac{199}{200} \iff q = \frac{1-e}{1+e} = \frac{1}{399}. \quad (5.2)$$

5.2 The endpoint derivative

Since $\Phi(0) = 0$, no positive uniform lower bound for Φ is possible on a closed interval beginning at zero. We instead use

$$J(e) = \frac{\Phi(e)}{e} \quad (e > 0), \quad J(0) = \Phi'(0). \quad (5.3)$$

Exact differentiation of the one-dimensional formula gives

$$\Phi'(0) = \frac{1}{2} \left[\frac{5\pi}{3\sqrt{3}} - 3 + \arccos \frac{1}{3} \left(\frac{3\sqrt{2}}{2} - \frac{5\sqrt{6}\pi}{18} \right) \right]. \quad (5.4)$$

Numerically,

$$\Phi'(0) = 0.0014901042070382105556 \dots > 0.$$

The endpoint calculation is expanded in [Section D](#).

5.3 Local endpoint certificate

On

$$0 \leq e \leq \frac{1}{100}, \quad \frac{99}{101} \leq q \leq 1,$$

the certificate encloses the integrated derivative and proves

$$\Phi'(e) > \frac{1}{2000}. \quad (5.5)$$

The domain is divided into sixteen rational q -slabs and ten rational x -panels per slab, for 160 terminal boxes. The weakest raw lower endpoint is

$$0.0005415852980838714162751786042230 \dots \quad (5.6)$$

Integrating (5.5) from zero to e gives

$$\Phi(e) \geq \frac{e}{2000}, \quad J(e) \geq \frac{1}{2000} \quad \left(0 \leq e \leq \frac{1}{100}\right), \quad (5.7)$$

where the endpoint $e = 0$ is interpreted by (5.3) and (5.4).

5.4 Compact bulk certificate

On

$$\frac{1}{100} \leq e \leq \frac{199}{200},$$

the proof encloses $\Phi(e)$ directly. The initial exact partition contains eight rational segments; only inconclusive segments are bisected. The final partition has 43 exact rational parameter slabs, ten x -panels per slab, 430 terminal boxes, and maximum adaptive depth eight.

Each slab proves the simple rational conclusion

$$\Phi(e) > \frac{1}{10^7}. \quad (5.8)$$

The weakest raw lower endpoint is

$$0.0000003289951660397968513639698874811370840941 \dots \quad (5.9)$$

on the first accepted bulk slab

$$\left[\frac{1}{100}, \frac{4293}{409600} \right].$$

The interval integration mechanism is described in [Section F](#). In short, a composite midpoint enclosure controls the x -integral, while a centered mean-value form controls the entire q -slab simultaneously.

5.5 Tail certificate

The final range

$$\frac{199}{200} \leq e < 1$$

corresponds to

$$0 < q \leq \frac{1}{399}.$$

Table 1: Summary of the scalar certificates.

Block	Parameter range	Certified statement	Precision	Boxes
Local	$0 \leq e \leq 1/100$	$\Phi(e)/e \geq 1/2000$	80 decimal digits	160
Bulk	$1/100 \leq e \leq 199/200$	$\Phi(e) > 10^{-7}$	80 decimal digits	430
Tail	$199/200 \leq e < 1$	$\Phi(e) > 1/20000$	80 decimal digits	2,000
Total	$0 < e < 1$	$\Phi(e) > 0$	–	2,590

The stable formula (4.36) extends continuously to $q = 0$, so the compact rectangle

$$0 \leq q \leq \frac{1}{399}, \quad 0 \leq x \leq 1$$

can be enclosed directly. The tail certificate uses five rational q -slabs and 400 rational x -panels per slab, for 2,000 terminal rectangles. It proves

$$\Phi(e) > \frac{1}{20000} \quad \left(\frac{199}{200} \leq e < 1 \right). \quad (5.10)$$

The weakest raw lower endpoint is

$$0.00006054637532642796468010985596478515 \dots \quad (5.11)$$

5.6 Combined scalar theorem

Theorem 5.1 (Certified scalar positivity). *The function Φ defined by (3.14) satisfies*

$$\Phi(0) = 0, \quad \Phi(e) > 0 \quad (0 < e < 1). \quad (5.12)$$

Proof. The local certificate gives positivity on $(0, 1/100]$ by (5.7). The bulk certificate gives positivity on $[1/100, 199/200]$ by (5.8). The tail certificate gives positivity on $[199/200, 1)$ by (5.10). The endpoints meet exactly, so the union covers every $0 < e < 1$. The identity $\Phi(0) = 0$ follows from the definition (3.14). $\square$

6 Completion of the volume theorem and equality cases

We now combine the exact additive reduction with the certified scalar theorem.

Proof of Theorem 1.1. By Theorem 3.3,

$$\text{Vol } K(\mathbf{e}, \sigma) - V_M = \Phi(e_1) + \Phi(e_2) + \Phi(e_3).$$

By Theorem 5.1, each summand is nonnegative. Hence

$$\text{Vol } K(\mathbf{e}, \sigma) \geq V_M.$$

If equality holds, then a sum of three nonnegative numbers is zero. Therefore

$$\Phi(e_1) = \Phi(e_2) = \Phi(e_3) = 0.$$

The scalar theorem says that $\Phi(e) = 0$ only at $e = 0$, so

$$e_1 = e_2 = e_3 = 0.$$

Conversely, the all-zero triple is the circle-line degeneration on every opposite-edge pair. It is a classical Meissner body and has volume V_M . This proves both the inequality and the equality statement. $\square$

At the all-zero parameter triple, the eight orientation patterns collapse under tetrahedral symmetry to the two familiar noncongruent Meissner types: one with the three rounded edges meeting at a vertex, and the other with the three rounded edges surrounding a face [9, 11]. Both have the same surface area and volume.

Corollary 6.1 (No confocal Peabody counterexample). *No width-one regular-tetrahedron confocal Peabody has volume strictly below the Meissner value V_M .*

Remark 6.2 (What the theorem does not say). The conclusion is not that Φ is globally increasing, and no global positivity statement is made for Φ' . The interval proof establishes only the positivity needed for the volume comparison. More importantly, the theorem does not compare Meissner bodies with arbitrary constant-width bodies. The global three-dimensional Blaschke–Lebesgue problem remains open.

7 Reproducibility and the computer-assisted trust boundary

The proof separates exact mathematical derivation from finite machine verification.

7.1 Exact components

The following steps are exact and do not depend on floating-point quadrature:

- the source Peabody construction and constant-width theorem;
- the reverse Gauss-map identity (3.1);
- the normal-sphere partition (3.4);
- the additive surface and volume formulas (3.9) and (3.15);
- the canonical confocal chart, the differential identities, and the integral formulas;
- the rational parameter transformations and the exact seams at $e = 1/100$ and $e = 199/200$;
- the equality-case argument.

7.2 Proof-authority computation

The scalar certificates use:

- exact integers and rational numbers for partition endpoints and control flow;
- `mpmath 1.3.0` interval arithmetic at 80 decimal digits;
- outward-rounded interval operations $+$, $-$, $\times$, $/$, $\sqrt{}$, and atan2 ;
- the midpoint Peano-kernel enclosure and a centered mean-value enclosure;
- 2,590 terminal interval boxes in total.

The central computation was independently replayed at 100 decimal digits and with reversed subdivision order. The tail computation was replayed at 60, 80, and 100 decimal digits. The accepted partitions, classifications, and stable leading lower-bound digits agreed.

7.3 Adversarial validation

The archived audit includes, among other checks:

1. high-precision agreement between original and stable formulas;
2. exact recovery of $\Phi(0) = 0$ and $\Phi'(0)$;
3. sampled quadrature containment inside certified interval slabs;
4. independent automatic-differentiation comparisons;
5. a deliberate sign mutation in the correction sector, which was detected;
6. a shifted $\Psi(0)$ baseline, which destroyed the certificate as expected;
7. an under-resolved partition, which was rejected;
8. exact seam and gap checks;
9. deterministic clean extraction and replay;
10. SHA-256 verification of every proof artifact.

The frozen theorem archive has SHA-256

b07df7201d2c5dcfaee65fa90c85a643f2bd55e81ce56fbf924b20e04e632459,

and the deterministic proof-core replay hash is

8ab4bfbf764da3c76a76810904e8289292b4c538957af68b15567a935882ac35.

7.4 Independent release audit

The intended release audit is deliberately separate from the principal certifier. It includes:

- an independent rederivation and implementation of the one-pair formula;
- a line-by-line audit of the width-two to width-one normalization;
- an independent review of the three-pair normal-sphere decomposition;
- fresh-kernel Mathematica replays with recursive unresolved-expression guards;
- a second certificate checker that does not import the principal certifier;
- a matching R semantic audit of formulas, orientations, partitions, metadata, and sample containment.

Mathematica and R are independent semantic checks, not the proof authority. This draft should be treated as a working manuscript until those release-audit gates and an independent human proofread of the geometric derivation are complete.

8 Discussion, scope, and non-extensions

The theorem closes the regular-tetrahedron confocal Peabody family as a source of a counterexample to Meissner minimality. It also supplies a structural explanation: after normal-sphere elimination, each pair of opposite edges contributes independently, and every nonzero contribution is positive.

This result should be viewed in relation to several narrower minimality theorems. Bogosel reduces the study of Meissner polyhedra to explicit dual-edge contributions and proves Meissner-tetrahedron minimality among Meissner pyramids [5]. The present theorem is different: it concerns a continuous confocal deformation family rather than the combinatorial class of Meissner pyramids. Both results support the Meissner conjecture without proving it globally.

The theorem also complements numerical and analytic comparisons with other tetrahedrally structured bodies. Robert’s body is a Peabody and is known numerically to have volume above the Meissner value; the present argument proves that the entire regular confocal path, including asymmetric three-parameter choices, remains above Meissner. The tetrahedrally symmetric body U_3 constructed by Arman, Bondarenko, Prymak, and Radchenko is likewise slightly larger than Meissner but lies outside the present confocal family [3].

8.1 Why no semi-regular extension is claimed

The Peabody source proves that a pair of confocal pea-pod devices can be fitted to the opposite edges of a semi-regular tetrahedron. That pairwise statement is not automatically a global construction theorem. The six-patch assembly used here is proved in the source for the three opposite-edge pairs of a *regular* tetrahedron, where the cap boundaries close compatibly and the three binormal families occupy the required global configuration.

For a deformed tetrahedral base, one would first need a new theorem showing that three independently chosen pairwise devices can be assembled simultaneously into six compatible wedge-pod patches and four closing caps, with global convexity and constant width. No such theorem is assumed here. Accordingly:

- no semi-regular numerical search is begun;
- no continuation of Φ beyond the regular-tetrahedron family is suggested;
- no claim is made for arbitrary Peabodies arising from self-dual ball polyhedra.

8.2 Relation to the global problem

The global Meissner conjecture asks whether the two Meissner bodies minimize volume among *all* three-dimensional bodies of a fixed constant width. That problem remains open [9, 1, 13]. The present theorem supplies a rigorous restricted-family minimum, not a universal lower bound. Its main value is to remove a substantial explicit family from the counterexample landscape and to expose an exact local-volume decomposition that may be useful elsewhere.

A Background for readers with multivariable calculus

This appendix collects the geometric facts used in the main proof and emphasizes the scaling and normalization steps.

A.1 Supporting planes and width

For a convex body $K \subset \mathbb{R}^3$ and a unit vector u , the support function is

$$h_K(u) = \max_{x \in K} x \cdot u.$$

The two supporting planes orthogonal to u have equations

$$x \cdot u = h_K(u), \quad x \cdot u = -h_K(-u).$$

Their distance is

$$h_K(u) + h_K(-u).$$

Thus K has constant width d exactly when

$$h_K(u) + h_K(-u) = d \quad \text{for every } u \in \mathbb{S}^2.$$

This is the three-dimensional analogue of a plane curve rolling between two parallel lines without changing their separation [6, 11].

A.2 Why opposite support points differ by the width

Let x and y be the support points in the directions u and $-u$. Then

$$(x - y) \cdot u = d.$$

A body of constant width d has diameter d , so $|x - y| \leq d$. Since

$$(x - y) \cdot u \leq |x - y||u| = |x - y|,$$

we have

$$d \leq |x - y| \leq d.$$

Equality in Cauchy–Schwarz forces $x - y = du$. This gives the reverse Gauss-map identity used in [Theorem 3.1](#).

A.3 Area scaling on a sphere

A sphere of radius r is obtained from the unit sphere by multiplying all lengths by r . Areas therefore scale by r^2 . For the Peabody caps, $r = 2$, so a cap has physical area four times the area of its Gauss image. This is the origin of the factor four in (3.6).

A.4 Blaschke’s identity and width normalization

For a three-dimensional body of constant width d , Blaschke’s identity is

$$V = \frac{d}{2}S - \frac{\pi}{3}d^3.$$

At width two,

$$V_2 = S_2 - \frac{8\pi}{3}.$$

Once this formula has produced a *volume*, scaling the body by $1/2$ multiplies the result by

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8}.$$

No extra area factor appears at that point. The line-by-line calculation is

$$S_2 = 8\pi + \sum_{i=1}^3 \Psi(e_i),$$

$$V_2 = S_2 - \frac{8\pi}{3} = \frac{16\pi}{3} + \sum_{i=1}^3 \Psi(e_i),$$

$$V_1 = \frac{1}{8}V_2 = \frac{2\pi}{3} + \frac{1}{8} \sum_{i=1}^3 \Psi(e_i).$$

At $e_1 = e_2 = e_3 = 0$,

$$V_M = \frac{2\pi}{3} + \frac{3\Psi(0)}{8}.$$

Subtracting the baseline gives

$$V_1 - V_M = \sum_{i=1}^3 \frac{\Psi(e_i) - \Psi(0)}{8}.$$

This is the exact reason for the definition $\Phi = (\Psi - \Psi(0))/8$.

B The regular-tetrahedron confocal chart in coordinates

We derive the parameter formula (4.2) and verify the local confocal identity with elementary algebra.

B.1 Beam endpoints

At the elliptic endpoint $t = \theta$, the condition $b \cos \theta = 1$ gives

$$X(\theta) = a \sin \theta \mathbf{k} + \mathbf{p}.$$

At the hyperbolic endpoint $\xi = \eta$, the condition $b \sinh \eta = 1$ gives

$$Y(\eta) = ae \cosh \eta \mathbf{k} + \mathbf{q}.$$

These are vertices of the regular tetrahedron, so their distance is two. Orthogonality of the frame gives

$$|X(\theta) - Y(\eta)|^2 = a^2(u_0 - ev_0)^2 + 1 + 1 = 4.$$

With the chosen orientation,

$$a(u_0 - ev_0) = \sqrt{2}. \quad (\text{B.1})$$

Since

$$u_0 = \frac{\sqrt{b^2 - 1}}{b}, \quad v_0 = \frac{\sqrt{b^2 + 1}}{b}, \quad a = \frac{b}{\sqrt{1 - e^2}},$$

Equation (B.1) becomes

$$\sqrt{b^2 - 1} - e\sqrt{b^2 + 1} = \sqrt{2(1 - e^2)}. \quad (\text{B.2})$$

A convenient algebraic parameter is

$$\tau = \frac{e^2}{1 - e^2}.$$

Writing $B^2 = b^2$, squaring (B.2) twice reduces it to

$$(B^2)^2 - (6 + 12\tau)B^2 + (9 + 4\tau + 4\tau^2) = 0.$$

Therefore

$$B^2 = 3 + 6\tau \pm 4\sqrt{2\tau(1 + \tau)}.$$

The minus sign gives the wrong confocal orientation; the admissible root is

$$B^2 = 3 + 6\tau + 4\sqrt{2\tau(1 + \tau)}.$$

Substituting $\tau = e^2/(1 - e^2)$ and

$$\sqrt{\tau(1 + \tau)} = \frac{e}{1 - e^2}$$

yields

$$b^2 = \frac{3 + 3e^2 + 4\sqrt{2}e}{1 - e^2}.$$

B.2 The width constant

We also need $a(v_0 - eu_0) = 2$. Let $\beta = b^2$. Then

$$a(v_0 - eu_0) = \frac{\sqrt{\beta + 1} - e\sqrt{\beta - 1}}{\sqrt{1 - e^2}}.$$

The formula for β gives the perfect-square identities

$$\beta + 1 = \frac{2(\sqrt{2} + e)^2}{1 - e^2}, \quad \beta - 1 = \frac{2(1 + \sqrt{2}e)^2}{1 - e^2}.$$

Hence

$$\begin{aligned} \sqrt{\beta + 1} - e\sqrt{\beta - 1} &= \frac{2 + \sqrt{2}e - e(1 + \sqrt{2}e)}{\sqrt{1 - e^2}} \\ &= 2\sqrt{1 - e^2}. \end{aligned}$$

Dividing by $\sqrt{1 - e^2}$ proves $a(v_0 - eu_0) = 2$.

B.3 Distance factorization

Using $b^2 = a^2(1 - e^2)$,

$$\begin{aligned} |X - Y|^2 &= a^2(\sin t - e \cosh \xi)^2 + b^2 \cos^2 t + b^2 \sinh^2 \xi \\ &= a^2(\cosh \xi - e \sin t)^2. \end{aligned}$$

On the parameter rectangle, $\cosh \xi - e \sin t > 0$, so

$$d = a(\cosh \xi - e \sin t).$$

Combining this with the radii (4.8) gives

$$\begin{aligned} d + R_E + R_H &= a(\cosh \xi - e \sin t) + ae(\sin t - u_0) + a(v_0 - \cosh \xi) \\ &= a(v_0 - eu_0) = 2. \end{aligned}$$

C Differential identities for the Gauss chart

Let

$$s = \sqrt{1 - e^2}, \quad D = \cosh \xi - e \sin t.$$

In the frame $(\mathbf{k}, \mathbf{p}, \mathbf{q})$, the normal map is

$$n = \frac{1}{D} \begin{pmatrix} \sin t - e \cosh \xi \\ s \cos t \\ -s \sinh \xi \end{pmatrix}.$$

The numerator has squared norm D^2 , so $|n| = 1$.

Direct differentiation gives

$$n_t = \frac{1}{D^2} \begin{pmatrix} (1 - e^2) \cos t \cosh \xi \\ s(e - \sin t \cosh \xi) \\ -es \cos t \sinh \xi \end{pmatrix},$$

$$n_\xi = \frac{1}{D^2} \begin{pmatrix} -(1-e^2) \sin t \sinh \xi \\ -s \cos t \sinh \xi \\ s(e \sin t \cosh \xi - 1) \end{pmatrix}.$$

Expanding the dot product and using $\cosh^2 \xi - \sinh^2 \xi = 1$ and $\sin^2 t + \cos^2 t = 1$ gives

$$n_t \cdot n_\xi = 0.$$

The same identities yield

$$|n_t|^2 = |n_\xi|^2 = \frac{1-e^2}{D^2}.$$

Since $b = a\sqrt{1-e^2}$ and $d = aD$, this is exactly

$$|n_t| = |n_\xi| = \frac{b}{d}.$$

Therefore

$$|n_t \times n_\xi| = \frac{b^2}{d^2},$$

which proves (4.13).

For the physical wedge surfaces, differentiate

$$U_E = X + R_E n, \quad U_H = Y - R_H n.$$

Because $X - Y = dn$, differentiation in t gives

$$X_t = d_t n + dn_t,$$

while $Y_t = 0$. Differentiating $d + R_E + R_H = 2$ in t and using $(R_H)_t = 0$ gives $d_t + (R_E)_t = 0$. Hence

$$(U_E)_t = (d + R_E)n_t = (2 - R_H)n_t.$$

Similarly, $(U_E)_\xi = R_E n_\xi$. The two hyperbolic identities follow in the same way. Orthogonality of n_t and n_ξ then gives the area elements in (4.17)–(4.18).

D Details of the one-dimensional reduction and endpoint derivative

D.1 The elementary kernel

For fixed t , put $C = e \sin t$ and

$$I(C) = \int_{-\eta}^{\eta} \frac{d\xi}{\cosh \xi - C}.$$

Use the half-angle substitution

$$z = \tanh \frac{\xi}{2}, \quad \cosh \xi = \frac{1+z^2}{1-z^2}, \quad d\xi = \frac{2 dz}{1-z^2}.$$

If $y = \tanh(\eta/2)$, then

$$I(C) = 2 \int_{-y}^y \frac{dz}{(1-C) + (1+C)z^2}$$

$$= \frac{4}{\sqrt{1-C^2}} \arctan \left(y \sqrt{\frac{1+C}{1-C}} \right).$$

Differentiation under the integral sign gives

$$I'(C) = \int_{-\eta}^{\eta} \frac{d\xi}{(\cosh \xi - C)^2}.$$

Since

$$v_0 - \cosh \xi = (v_0 - C) - (\cosh \xi - C),$$

we obtain

$$\int_{-\eta}^{\eta} \frac{v_0 - \cosh \xi}{(\cosh \xi - C)^2} d\xi = (v_0 - C)I'(C) - I(C).$$

Differentiating the explicit arctangent expression and using $\sinh \eta = 1/b$ simplifies the right-hand side to

$$(v_0 - C)I'(C) - I(C) = \frac{(v_0 C - 1)I(C) + 2/b}{1 - C^2}. \quad (\text{D.1})$$

Substitution into (4.20), followed by the symmetry $\sin(\pi - t) = \sin t$, yields (4.22).

D.2 The Meissner endpoint

At $e = 0$,

$$a = b = \sqrt{3}, \quad \theta = \arccos \frac{1}{\sqrt{3}}, \quad \eta = \operatorname{arsinh} \frac{1}{\sqrt{3}}.$$

Moreover,

$$y = \tanh \frac{\eta}{2} = 2 - \sqrt{3},$$

and

$$I(0) = 4 \arctan(2 - \sqrt{3}) = \frac{\pi}{3}.$$

The integrand in (4.22) becomes the constant $I(0)/a = \pi/(3\sqrt{3})$. Since

$$\pi - 2\theta = \arccos \frac{1}{3},$$

we recover

$$\Psi(0) = -4(3) \left(\frac{\pi}{2} - \theta \right) \frac{\pi}{3\sqrt{3}} = -\frac{2\pi}{\sqrt{3}} \arccos \frac{1}{3}.$$

D.3 Right derivative at zero

The moving-limit Leibniz rule is

$$\frac{d}{de} \int_{\theta(e)}^{\pi/2} F(e, t) dt = -F(e, \theta(e))\theta'(e) + \int_{\theta(e)}^{\pi/2} F_e(e, t) dt.$$

The needed endpoint values are

$$b(0) = a(0) = \sqrt{3}, \quad b'(0) = a'(0) = \frac{2\sqrt{6}}{3}, \quad \theta'(0) = \frac{2}{3},$$

$$I(0) = \frac{\pi}{3}, \quad I_C(0) = 1.$$

Substitution into the derivative of (4.22), followed by elementary integration of 1, $\sin t$, and the endpoint constants, gives

$$\Psi'(0) = 4 \left[\frac{5\pi}{3\sqrt{3}} - 3 + \arccos \frac{1}{3} \left(\frac{3\sqrt{2}}{2} - \frac{5\sqrt{6}\pi}{18} \right) \right]. \quad (\text{D.2})$$

Since $\Phi = (\Psi - \Psi(0))/8$, division by eight gives (5.4).

E Algebra and branch control in the stable q -chart

The parameter

$$q = \frac{1 - e}{1 + e}$$

is equivalent to the rapidity coordinate $e = \tanh \lambda$, since $q = e^{-2\lambda}$. Put

$$\kappa = 1 + \sqrt{2}, \quad K = \kappa^2 = 3 + 2\sqrt{2}.$$

Then

$$1 - e^2 = \frac{4q}{(1 + q)^2}.$$

Substitution into (4.2) gives the exact identities

$$b^2 = \frac{K^2 + q^2}{2Kq}, \quad a^2 = \frac{(K^2 + q^2)(1 + q)^2}{8Kq^2}. \quad (\text{E.1})$$

Equivalently, if

$$\rho = \kappa e^\lambda = \frac{\kappa}{\sqrt{q}},$$

then

$$b^2 = \frac{\rho^4 + 1}{2\rho^2}.$$

This is the compact exponential-rapidity form used during the symbolic reduction.

The beam substitution $x = b \cos t$ introduces expressions in which $R - (K - q)$ would suffer cancellation near $q = 0$. The stable formula rationalizes that difference:

$$\begin{aligned} R - (K - q) &= \frac{R^2 - (K - q)^2}{R + K - q} \\ &= q \frac{2K(1 - x^2)}{R + K - q} = q\delta. \end{aligned}$$

This explains the definition of δ in (4.30). After the same rationalization is applied to the remaining terms, the fixed-interval integrand becomes (4.36).

E.1 No branch ambiguity

For $0 \leq q \leq 1$ and $0 \leq x \leq 1$,

$$\begin{aligned} D^2 &= K^2 + q^2 > 0, \\ R^2 &= K^2 + q^2 - 2Kqx^2 \geq (K - q)^2 > 0, \end{aligned}$$

and

$$T^2 = 2K(K^2 + q^2) + K^2(1 - q)^2 x^2 > 0.$$

Since $K > 1$,

$$R + K - q \geq 2(K - q) > 0, \quad K + q + D > 0.$$

Thus every radical is the positive real branch and every denominator in (4.27)–(4.35) is strictly positive. The argument of arctan is nonnegative, so the principal real branch is continuous across the full certificate rectangle. These inequalities are also checked by outward interval arithmetic in the archived domain diagnostics.

F How the interval certificates work

This appendix explains the finite verification without assuming prior experience with interval arithmetic. A standard reference is [12].

F.1 Outward-rounded intervals

Instead of storing a real number x , interval arithmetic stores a guaranteed enclosure

$$X = [\underline{X}, \overline{X}] \quad \text{with} \quad x \in X.$$

An interval operation is rounded outward. For example, if $x \in X$ and $y \in Y$, the computed interval $X + Y$ is guaranteed to contain every $x + y$. The same principle applies to multiplication, division when zero is excluded, square roots of positive intervals, and inverse trigonometric functions.

The benefit is logical: if the lower endpoint of a computed interval is positive, then every real value represented by that interval is positive. The cost is dependency inflation when the same variable appears many times. The certificate controls that inflation by using stable algebra, small rational parameter slabs, midpoint quadrature in x , and centered mean-value forms in q .

F.2 Midpoint enclosure for one panel

Let $[a, b]$ have width $h = b - a$ and midpoint $m = (a + b)/2$. Taylor's theorem with integral remainder implies

$$\int_a^b f(x) \, dx = hf(m) + R,$$

where the midpoint Peano kernel is nonnegative and has total mass $h^3/24$. If an interval computation gives

$$f''(x) \in [L, U] \quad (a \leq x \leq b),$$

then

$$\int_a^b f(x) \, dx \in hf(m) + \frac{h^3}{24}[L, U]. \tag{F.1}$$

The central certifier uses a nested automatic-differentiation jet to evaluate

$$H, \quad H_q, \quad H_{xx}, \quad H_{qxx}$$

as outward intervals.

F.3 One enclosure for an entire parameter slab

Let $Q = [q_-, q_+]$ have midpoint q_m . The ordinary mean-value theorem gives

$$\Psi(q) = \Psi(q_m) + (q - q_m)\Psi_q(\xi)$$

for some $\xi \in Q$. Therefore an interval enclosure of $\Psi_q(Q)$ implies

$$\Psi(Q) \subseteq \Psi(q_m) + (Q - q_m)\Psi_q(Q). \quad (\text{F.2})$$

Both $\Psi(q_m)$ and $\Psi_q(Q)$ are obtained by summing panel enclosures of the form (F.1). Subtracting an independent enclosure of $\Psi(0)$ and dividing by eight gives one interval containing $\Phi(e)$ for every parameter in the slab.

F.4 The local endpoint

On the local range, the certifier encloses Ψ_q . Since

$$\frac{dq}{de} = -\frac{2}{(1+e)^2} = -\frac{(1+q)^2}{2},$$

we have

$$\Phi'(e) = \frac{1}{8}\Psi_q(q)\frac{dq}{de}.$$

The interval product proves $\Phi'(e) > 1/2000$ on each local slab. The fundamental theorem of calculus then converts the derivative bound into the quotient bound for $J(e) = \Phi(e)/e$.

F.5 The tail

Near $q = 0$, the stable integrand extends continuously and the direct volume increment is comfortably separated from zero. The tail proof therefore evaluates $H(Q, X)$ on each of 2,000 rational rectangles and sums the signed lower bounds over the x -panels. It certifies Φ directly rather than certifying Φ' .

F.6 Why the integrand need not be positive

The exact derivative integrand changes sign. A proof that required every pointwise contribution to be positive would therefore target a false intermediate statement. Interval integration is used on the signed contributions, and the final summed lower endpoint is the proof target. This is why panelwise integration and cancellation control are essential.

G Complete certificate slab tables

The tables below are generated directly from the frozen CSV summaries. Lower bounds are rounded for display only; the certificate stores longer outward-rounded decimal endpoints and exact rational partition endpoints.

Table 2: Exact local endpoint slabs and certified lower bounds for $\Phi'(e)$.

Index	e interval	q interval	lower bound for $\Phi'(e)$
15	$[0, \frac{1}{1615}]$	$[\frac{807}{808}, 1]$	$5.679735807188383 \times 10^{-4}$
14	$[\frac{1}{1615}, \frac{1}{807}]$	$[\frac{404}{808}, \frac{807}{808}]$	$5.661986974652423 \times 10^{-4}$
13	$[\frac{1}{807}, \frac{1}{1613}]$	$[\frac{805}{808}, \frac{404}{808}]$	$5.644257524388871 \times 10^{-4}$
12	$[\frac{3}{1613}, \frac{1}{403}]$	$[\frac{201}{808}, \frac{805}{808}]$	$5.626550916029019 \times 10^{-4}$
11	$[\frac{1}{403}, \frac{5}{1611}]$	$[\frac{803}{808}, \frac{201}{808}]$	$5.608867058953334 \times 10^{-4}$
10	$[\frac{1}{1611}, \frac{805}{808}]$	$[\frac{404}{808}, \frac{803}{808}]$	$5.591205963517944 \times 10^{-4}$
9	$[\frac{3}{805}, \frac{7}{1609}]$	$[\frac{801}{808}, \frac{401}{808}]$	$5.573567749966684 \times 10^{-4}$
8	$[\frac{7}{1609}, \frac{1}{201}]$	$[\frac{100}{808}, \frac{801}{808}]$	$5.555952431160704 \times 10^{-4}$
7	$[\frac{1}{201}, \frac{1}{1607}]$	$[\frac{799}{808}, \frac{101}{808}]$	$5.538360019883969 \times 10^{-4}$
6	$[\frac{9}{1607}, \frac{5}{803}]$	$[\frac{399}{808}, \frac{799}{808}]$	$5.520790528843305 \times 10^{-4}$
5	$[\frac{5}{803}, \frac{11}{1605}]$	$[\frac{797}{808}, \frac{399}{808}]$	$5.503243970668435 \times 10^{-4}$
4	$[\frac{11}{1605}, \frac{401}{1603}]$	$[\frac{808}{808}, \frac{404}{808}]$	$5.485720014138190 \times 10^{-4}$
3	$[\frac{3}{401}, \frac{13}{1603}]$	$[\frac{795}{808}, \frac{199}{808}]$	$5.468218784111319 \times 10^{-4}$
2	$[\frac{13}{1603}, \frac{801}{801}]$	$[\frac{397}{808}, \frac{795}{808}]$	$5.450740527020893 \times 10^{-4}$
1	$[\frac{7}{801}, \frac{15}{1601}]$	$[\frac{797}{808}, \frac{397}{808}]$	$5.433285255183472 \times 10^{-4}$
0	$[\frac{15}{1601}, \frac{1}{100}]$	$[\frac{99}{808}, \frac{793}{808}]$	$5.415852980838714 \times 10^{-4}$

Table 3: Adaptive compact-bulk partition and certified lower bounds for $\Phi(e)$.

Index	depth	e interval	lower bound for $\Phi(e)$
0	8	$[\frac{1}{100}, \frac{4293}{409600}]$	$3.289951660397969 \times 10^{-7}$
1	8	$[\frac{4293}{409600}, \frac{449}{409600}]$	$1.027220841708625 \times 10^{-6}$
2	7	$[\frac{449}{409600}, \frac{1221}{102400}]$	$1.277730552757676 \times 10^{-6}$
3	6	$[\frac{1221}{102400}, \frac{51200}{51200}]$	$9.598744664355846 \times 10^{-7}$
4	6	$[\frac{709}{51200}, \frac{323}{20480}]$	$3.745526402742635 \times 10^{-6}$
5	6	$[\frac{323}{20480}, \frac{453}{25600}]$	$6.511122438753733 \times 10^{-6}$
6	5	$[\frac{453}{25600}, \frac{1103}{51200}]$	$2.759003222474406 \times 10^{-6}$
7	5	$[\frac{1103}{51200}, \frac{13}{512}]$	$8.310654012050498 \times 10^{-6}$
8	5	$[\frac{13}{512}, \frac{1497}{51200}]$	$1.378161027059568 \times 10^{-5}$
9	5	$[\frac{1497}{51200}, \frac{847}{25600}]$	$1.917314227180660 \times 10^{-5}$
10	4	$[\frac{847}{25600}, \frac{261}{6400}]$	$7.846349019286869 \times 10^{-7}$
11	4	$[\frac{261}{6400}, \frac{1241}{25600}]$	$1.202293004009885 \times 10^{-5}$
12	4	$[\frac{1241}{25600}, \frac{719}{12800}]$	$2.292710511216354 \times 10^{-5}$
13	4	$[\frac{719}{12800}, \frac{327}{5120}]$	$3.350799803457227 \times 10^{-5}$
14	4	$[\frac{327}{5120}, \frac{229}{3200}]$	$4.377619132553730 \times 10^{-5}$
15	4	$[\frac{229}{3200}, \frac{2029}{25600}]$	$5.374174922146528 \times 10^{-5}$
16	4	$[\frac{2029}{25600}, \frac{1113}{12800}]$	$6.341466118558133 \times 10^{-5}$
17	3	$[\frac{1113}{12800}, \frac{131}{1280}]$	$3.665552477877724 \times 10^{-7}$
18	3	$[\frac{131}{1280}, \frac{1507}{12800}]$	$2.307173806690166 \times 10^{-5}$
19	3	$[\frac{1507}{12800}, \frac{213}{1600}]$	$4.437070995191022 \times 10^{-5}$
20	3	$[\frac{213}{1600}, \frac{1901}{12800}]$	$6.436201220800802 \times 10^{-5}$
21	3	$[\frac{1901}{12800}, \frac{1049}{6400}]$	$8.313473105261038 \times 10^{-5}$
22	3	$[\frac{1049}{6400}, \frac{459}{2560}]$	$1.007665567684218 \times 10^{-4}$
23	3	$[\frac{459}{2560}, \frac{623}{3200}]$	$1.173360208142342 \times 10^{-4}$
24	3	$[\frac{623}{3200}, \frac{2689}{12800}]$	$1.329145881189513 \times 10^{-4}$
25	3	$[\frac{2689}{12800}, \frac{1243}{6400}]$	$1.475714537816976 \times 10^{-4}$
26	2	$[\frac{1243}{6400}, \frac{41}{160}]$	$3.086945255647257 \times 10^{-6}$
27	2	$[\frac{41}{160}, \frac{1837}{6400}]$	$4.562525070304942 \times 10^{-5}$
28	2	$[\frac{1837}{6400}, \frac{1017}{3200}]$	$8.303453120678563 \times 10^{-5}$
29	2	$[\frac{1017}{3200}, \frac{2231}{6400}]$	$1.160267563312698 \times 10^{-4}$
30	2	$[\frac{2231}{6400}, \frac{607}{1600}]$	$1.451553147180399 \times 10^{-4}$
31	2	$[\frac{607}{1600}, \frac{189}{256}]$	$1.709230636789312 \times 10^{-4}$
32	2	$[\frac{189}{256}, \frac{1411}{3200}]$	$1.937526799098876 \times 10^{-4}$
33	2	$[\frac{1411}{3200}, \frac{3019}{6400}]$	$2.139751856998294 \times 10^{-4}$
34	2	$[\frac{3019}{6400}, \frac{201}{400}]$	$2.319125224513017 \times 10^{-4}$
35	1	$[\frac{201}{400}, \frac{361}{640}]$	$1.260650910712992 \times 10^{-5}$
36	1	$[\frac{361}{640}, \frac{1001}{1600}]$	$7.779333893552551 \times 10^{-5}$
37	1	$[\frac{1001}{1600}, \frac{2199}{3200}]$	$1.295914618990274 \times 10^{-4}$

Index	depth	e interval	lower bound for $\Phi(e)$
38	1	$[\frac{2199}{3200}, \frac{599}{800}]$	$1.710106601166309 \times 10^{-4}$
39	1	$[\frac{399}{800}, \frac{2593}{3200}]$	$2.041162859471716 \times 10^{-4}$
40	1	$[\frac{2593}{3200}, \frac{279}{320}]$	$2.306792283181848 \times 10^{-4}$
41	1	$[\frac{279}{320}, \frac{2987}{3200}]$	$2.519003547908074 \times 10^{-4}$
42	1	$[\frac{2987}{3200}, \frac{199}{200}]$	$2.687310006464714 \times 10^{-4}$

Table 4: Tail slabs in the stable q chart and certified lower bounds for Φ .

Index	q interval	lower bound for $\Phi(e(q))$
0	$[0, \frac{1}{1995}]$	$6.085485507529240 \times 10^{-5}$
1	$[\frac{1}{1995}, \frac{2}{1995}]$	$6.077797013128531 \times 10^{-5}$
2	$[\frac{2}{1995}, \frac{1}{665}]$	$6.070092850819833 \times 10^{-5}$
3	$[\frac{1}{665}, \frac{4}{1995}]$	$6.062373023143653 \times 10^{-5}$
4	$[\frac{4}{1995}, \frac{1}{399}]$	$6.054637532642796 \times 10^{-5}$

H Reproducibility record and build instructions

H.1 Frozen theorem package

The manuscript is based on the archive

`confocal_peabody_family_minimality_FROZEN_2026-09-30.zip`

with SHA-256

`b07df7201d2c5dcfaee65fa90c85a643`
`b2bd55e81ce56fbf924b20e04e632459`

The self-verifying archive contains the 50 manifest-listed theorem files together with the two root ledgers

`PACKAGE_MANIFEST.json` `SHA256SUMS.txt`.

The package validator reproduces the classification

`PEABODY_FAMILY_PACKAGE_VALIDATION_PASS.`

The deterministic proof-core hash is

`8ab4bfbf764da3c76a76810904e828929`
`2b4c538957af68b15567a935882ac35b`

H.2 Manuscript build

From the manuscript directory, regenerate the appendix tables and compile with

```
python3 tools/generate_certificate_tables.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

A clean rebuild is obtained with

```
latexmk -C
python3 tools/generate_certificate_tables.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

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